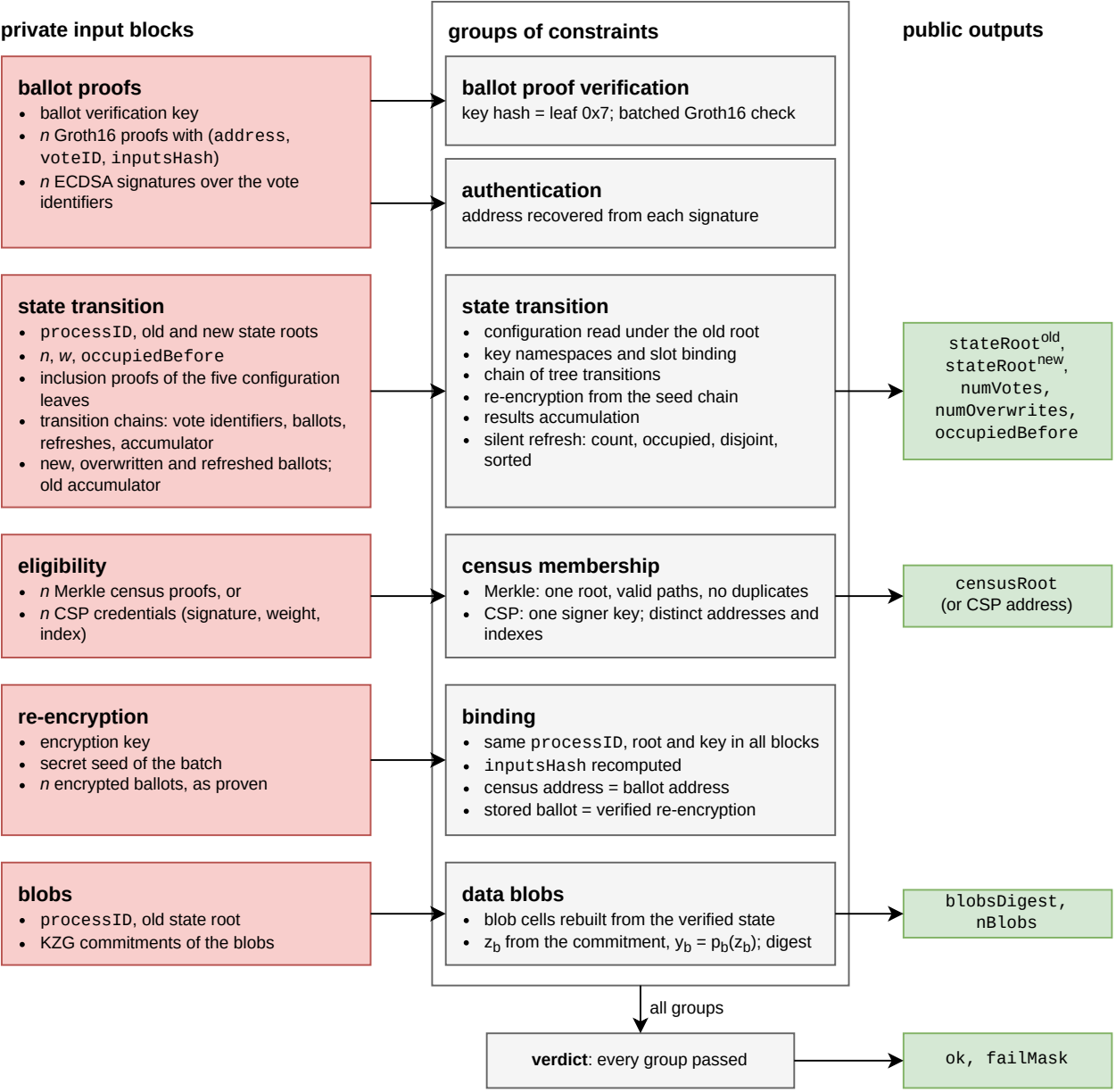
private input blocks
groups of constraints
public outputs
ballot proofs
ballot verification key
n Groth16 proofs with (address, voteID, inputsHash)
n ECDSA signatures over the vote identifiers
ballot proof verification
key hash = leaf 0x7; batched Groth16 check
authentication
address recovered from each signature
state transition
processID, old and new state roots
n, w, occupiedBefore
inclusion proofs of the five configuration leaves
transition chains: vote identifiers, ballots, refreshes, accumulator
new, overwritten and refreshed ballots; old accumulator
state transition
configuration read under the old root
key namespaces and slot binding
chain of tree transitions
re-encryption from the seed chain
results accumulation
silent refresh: count, occupied, disjoint, sorted
stateRoot^old, stateRoot^new, numVotes, numOverwrites, occupiedBefore
eligibility
n Merkle census proofs, or
n CSP credentials (signature, weight, index)
census membership
Merkle: one root, valid paths, no duplicates
CSP: one signer key; distinct addresses and indexes
censusRoot (or CSP address)
re-encryption
encryption key
secret seed of the batch
n encrypted ballots, as proven
binding
same processID, root and key in all blocks
inputsHash recomputed
census address = ballot address
stored ballot = verified re-encryption
blobs
processID, old state root
KZG commitments of the blobs
data blobs
blob cells rebuilt from the verified state
z_b from the commitment, y_b = p_b(z_b); digest
blobsDigest, nBlobs
all groups
verdict: every group passed
ok, failMask